Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–40

Alex Vergara

November 4, 2026

1. Consider the following matrix: $A = \begin{bmatrix} 11 & \frac{1}{2} & -2 & 3 \\ 4 & 1 & 1 & 0 \end{bmatrix}$.

   (a) What type of matrix is A? Is a 2 by 4 matrix (2×4), so it's a rectangular matrix

   (b) What are the entries of A at positions $(1, 2)$ and $(2, 1)$?

   - $a_{1,2} = \frac{1}{2}$
   - $a_{2,1} = 4$

   (c) At which positions in A is the entry equal to 1? In $a_{2,2}$ and $a_{2,3}$

2. Write the matrix $\left(A = [a_{ij}]_{3\times 3}\right)$ defined by:

   (a) $a_{ij} = i + j$

   $$A = \begin{pmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}$$

   (b) $a_{ij} = \begin{cases} 0, & \text{if } i = j, \\ -1, & \text{if } i > j, \\ 2, & \text{if } i < j. \end{cases}$

   $$A = \begin{pmatrix} 0 & 2 & 2 \\ -1 & 0 & 2 \\ -1 & -1 & 0 \end{pmatrix}$$

3. For each of the following parts, determine, if possible, the real numbers x and y such that the matrices A and B are equal.

   Remember: Two matrices are equal if they have the same dimensions and all their corresponding entries **are equal**.

   (a) $A = \begin{bmatrix} 1 & 0 \\ x & y \\ -1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ \frac{y}{2} & x+1 \\ -1 & 3 \end{bmatrix}$

   Solution:

- From the first row: $a_{11} = 1 = b_{11}$ and $a_{12} = 0 = b_{12}$.
- From the second row: $a_{21} = x = \frac{y}{2} = b_{21}$ and $a_{22} = y = x + 1 = b_{22}$.
- From the third row: $a_{31} = -1 = b_{31}$ and $a_{32} = 3 = b_{32}$.

The point is that rows 1 and 3 do not give us any information about x and y, so we only need to solve the system of equations from the second row:

$$\begin{cases} x = \dfrac{y}{2}, \\ y = x + 1. \end{cases}$$

Substitute $x = \frac{y}{2}$ into the second equation:

$$\begin{aligned} y &= \frac{y}{2} + 1, \\ 2y &= y + 2, \\ y &= 2, \qquad x = \frac{y}{2} = 1. \end{aligned}$$

$$x = 1, \qquad y = 2$$

(b) $A = \begin{bmatrix} x^3 & 2 & 0 \\ 3 & y & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 3x+1 & 2 & y \\ 3 & -2 & -1 \end{bmatrix}$

Solution:

(c) $A = \begin{bmatrix} 2 & -5 \\ 0 & x+y \\ 2y & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -5 \\ 0 & 2x+4 \\ y+1 & 1 \end{bmatrix}$

Solution:

4. Consider the following matrices:

$$A = \begin{bmatrix} 2 & 2 \\ 3 & 3 \\ -1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 3 \\ -4 & 0 \\ 0 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix},$$

$$D = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & -2 \end{bmatrix}, \quad E = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 2 & 1 \\ -2 & 2 & 4 \end{bmatrix}, \quad F = \begin{bmatrix} -2 & 1 & 3 \\ -1 & 1 & -2 \\ 0 & 1 & 3 \end{bmatrix}$$

Determine the following matrices, only if they are defined. If they are not defined, explain why.

A) $A + B$	E) $-A + 2B$	I) $C - C$	M) AD
B) $B + A$	F) $A + C$	J) AB	N) DA
C) $A - B$	G) $A + D$	K) AC	O) EF
D) $D + E$	H) $A - A$	L) CA	P) FE

Q) $A(3D)$ S) $A(CD)$ U) $(2A)((2C)(2D))$ W) C^3
R) $(AC)D$ T) $(2A)(C(3D))$ V) $A(CD+D)$

A) $\begin{bmatrix} 1 & 5 \\ -1 & 3 \\ -1 & -2 \end{bmatrix}$, B) $\begin{bmatrix} 1 & 5 \\ -1 & 3 \\ -1 & -2 \end{bmatrix}$ C) $\begin{bmatrix} 3 & -1 \\ 7 & 3 \\ -1 & 0 \end{bmatrix}$

D) Not defined, because D is a 2×3 matrix and E is a 3×3 matrix, so they do not have the same dimensions.

E) $-1 \cdot \begin{bmatrix} 2 & 2 \\ 3 & 3 \\ -1 & -1 \end{bmatrix} + 2 \cdot \begin{bmatrix} -1 & 3 \\ -4 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ -11 & -3 \\ 1 & -1 \end{bmatrix}$

F) Not defined, because A is a 3×2 matrix and C is a 2×2 matrix, so they do not have the same dimensions.

G) Not defined, because A is a 3×2 matrix and D is a 2×3 matrix, so they do not have the same dimensions.

H) $A - A = 0_{3\times 2}$

I) $C - C = 0_{2\times 2}$

J) $A, B \in \mathcal{M}_{3\times 2}(\mathbb{R})$. Since $2 \neq 3$, AB is undefined.

K) Given $A \in \mathcal{M}_{3\times 2}(\mathbb{R})$ and $C \in \mathcal{M}_{2\times 2}(\mathbb{R})$,

$$(AC)_{ij} = \sum_{k=1}^{2} a_{ik}c_{kj}, \quad i \in \{1,2,3\},\ j \in \{1,2\}.$$

$$(AC)_{11} = \sum_{k=1}^{2} a_{1k}c_{k1} = 2(2) + 2(3) = 10,$$

$$(AC)_{12} = \sum_{k=1}^{2} a_{1k}c_{k2} = 2(-1) + 2(0) = -2,$$

$$(AC)_{21} = \sum_{k=1}^{2} a_{2k}c_{k1} = 3(2) + 3(3) = 15,$$

$$(AC)_{22} = \sum_{k=1}^{2} a_{2k}c_{k2} = 3(-1) + 3(0) = -3,$$

$$(AC)_{31} = \sum_{k=1}^{2} a_{3k}c_{k1} = (-1)(2) + (-1)(3) = -5,$$

$$(AC)_{32} = \sum_{k=1}^{2} a_{3k}c_{k2} = (-1)(-1) + (-1)(0) = 1 \implies$$

$$\implies AC = \begin{bmatrix} 10 & -2 \\ 15 & -3 \\ -5 & 1 \end{bmatrix}$$

L) Given $C \in \mathcal{M}_{2\times 2}(\mathbb{R})$ and $A \in \mathcal{M}_{3\times 2}(\mathbb{R})$, the matrix product CA is undefined, because the number of columns of C is not equal to the number of rows of A.

M) Given $A \in \mathcal{M}_{3\times 2}(\mathbb{R})$ and $D \in \mathcal{M}_{2\times 3}(\mathbb{R})$,

$$(AD)_{ij} = \sum_{k=1}^{2} a_{ik} d_{kj}, \quad i \in \{1,2,3\},\ j \in \{1,2,3\}.$$

$$(AD)_{11} = \sum_{k=1}^{2} a_{1k} d_{k1} = 2(1) + 2(3) = 8,$$

$$(AD)_{12} = \sum_{k=1}^{2} a_{1k} d_{k2} = 2(2) + 2(1) = 6,$$

$$(AD)_{13} = \sum_{k=1}^{2} a_{1k} d_{k3} = 2(3) + 2(-2) = 2,$$

$$(AD)_{21} = \sum_{k=1}^{2} a_{2k} d_{k1} = 3(1) + 3(3) = 12,$$

$$(AD)_{22} = \sum_{k=1}^{2} a_{2k} d_{k2} = 3(2) + 3(1) = 9,$$

$$(AD)_{23} = \sum_{k=1}^{2} a_{2k} d_{k3} = 3(3) + 3(-2) = 3,$$

$$(AD)_{31} = \sum_{k=1}^{2} a_{3k} d_{k1} = (-1)(1) + (-1)(3) = -4,$$

$$(AD)_{32} = \sum_{k=1}^{2} a_{3k} d_{k2} = (-1)(2) + (-1)(1) = -3,$$

$$(AD)_{33} = \sum_{k=1}^{2} a_{3k} d_{k3} = (-1)(3) + (-1)(-2) = -1 \implies$$

$$\implies AD = \begin{bmatrix} 8 & 6 & 2 \\ 12 & 9 & 3 \\ -4 & -3 & -1 \end{bmatrix}$$

N) Given $D \in \mathcal{M}_{2\times 3}(\mathbb{R})$ and $A \in \mathcal{M}_{3\times 2}(\mathbb{R})$,

$$(DA)_{ij} = \sum_{k=1}^{3} d_{ik} a_{kj}, \quad i \in \{1,2\},\ j \in \{1,2\}.$$

$$(DA)_{11} = \sum_{k=1}^{3} d_{1k}a_{k1} = 1(2) + 2(3) + 3(-1) = 5,$$

$$(DA)_{12} = \sum_{k=1}^{3} d_{1k}a_{k2} = 1(2) + 2(3) + 3(-1) = 5,$$

$$(DA)_{21} = \sum_{k=1}^{3} d_{2k}a_{k1} = 3(2) + 1(3) + (-2)(-1) = 11,$$

$$(DA)_{22} = \sum_{k=1}^{3} d_{2k}a_{k2} = 3(2) + 1(3) + (-2)(-1) = 11 \implies$$

$$\implies DA = \begin{bmatrix} 5 & 5 \\ 11 & 11 \end{bmatrix}$$

O) Given $E \in \mathcal{M}_{3\times 3}(\mathbb{R})$ and $F \in \mathcal{M}_{3\times 3}(\mathbb{R})$,

$$(EF)_{ij} = \sum_{k=1}^{3} e_{ik}f_{kj}, \quad i \in \{1,2,3\},\ j \in \{1,2,3\} \implies$$

$$EF = \begin{bmatrix} -2 & 3 & 9 \\ 0 & 2 & -4 \\ 2 & 4 & 2 \end{bmatrix}$$

P) Given $F \in \mathcal{M}_{3\times 3}(\mathbb{R})$ and $E \in \mathcal{M}_{3\times 3}(\mathbb{R})$,

$$(FE)_{ij} = \sum_{k=1}^{3} f_{ik}e_{kj}, \quad i \in \{1,2,3\},\ j \in \{1,2,3\}.$$

$$FE = \begin{bmatrix} -9 & 8 & 9 \\ 2 & -2 & -9 \\ -7 & 8 & 13 \end{bmatrix}$$

Q) $A(3D)$. We know from the previous result on M that AD is defined, so $A(3D)$ is also defined. The unique change is that D is now multiplied by 3, and given the property of matrix multiplication:

$$\alpha(XY) = (\alpha X)Y = X(\alpha Y), \quad \alpha \in \mathbb{R},\ X \in \mathcal{M}_{m\times n}(\mathbb{R}),\ Y \in \mathcal{M}_{n\times p}(\mathbb{R}).$$

$\therefore$, we can write:

$$A(3D) = 3(AD) = 3 \cdot \begin{bmatrix} 8 & 6 & 2 \\ 12 & 9 & 3 \\ -4 & -3 & -1 \end{bmatrix} = \begin{bmatrix} 24 & 18 & 6 \\ 36 & 27 & 9 \\ -12 & -9 & -3 \end{bmatrix}$$

R) $(AC)D$. We know from the previous result on K that AC is defined. $AC \in \mathcal{M}_{3\times 2}(\mathbb{R})$ and $D \in \mathcal{M}_{2\times 3}(\mathbb{R}) \implies$ that $(AC) \cdot D$ is defined.

$$(AC)D = \begin{bmatrix} 10 & -2 \\ 15 & -3 \\ -5 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 4 & 18 & 34 \\ 6 & 27 & 51 \\ -2 & -9 & -17 \end{bmatrix}$$

S) CD is defined because C is 2×2 and D is 2×3; hence CD is 2×3 and $A(CD)$ is defined. By associativity of matrix multiplication and the result in R),

$$A(CD) = (AC)D = \begin{bmatrix} 4 & 18 & 34 \\ 6 & 27 & 51 \\ -2 & -9 & -17 \end{bmatrix}.$$

T) Let's call T to the matrix result of the evaluation of the expression in the line below.
By associativity of matrix multiplication and compatibility with scalar multiplication, we can regroup the factors and factor out the scalars:

$$T = (2A)(C(3D)) = 2 \cdot 3\big(A(CD)\big) = 6(AC)D$$

Since we know that $T = 6 \cdot (AC)D$, then:

$$\implies 6 \cdot \begin{bmatrix} 4 & 18 & 34 \\ 6 & 27 & 51 \\ -2 & -9 & -17 \end{bmatrix} = \begin{bmatrix} 24 & 108 & 204 \\ 36 & 162 & 306 \\ -12 & -54 & -102 \end{bmatrix}$$

U) By associativity and compatibility with scalar multiplication,

$$(2A)((2C)(2D)) = 8(AC)D = \begin{bmatrix} 32 & 144 & 272 \\ 48 & 216 & 408 \\ -16 & -72 & -136 \end{bmatrix}.$$

V) By distributivity and associativity, using R) and M),

$$A(CD + D) = (AC)D + AD = \begin{bmatrix} 12 & 24 & 36 \\ 18 & 36 & 54 \\ -6 & -12 & -18 \end{bmatrix}.$$

W)

$$C^2 = \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 6 & -3 \end{bmatrix}.$$

Therefore,

$$C^3 = C^2C = \begin{bmatrix} 1 & -2 \\ 6 & -3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} -4 & -1 \\ 3 & -6 \end{bmatrix}.$$

5. Consider the following matrices:

$$A = \begin{bmatrix} 0 & 3 & 1 \\ -1 & 0 & 2 \\ 3 & 0 & 2 \end{bmatrix} \quad \textit{and} \quad B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 3 \\ 2 & 8 & 1 \end{bmatrix}$$

Without computing the product, calculate:

a) $(AB)_{23} = (-1)\cdot 0 + 0\cdot 3 + 2\cdot 1 = 2$

$(AB)_{31} = 3(-1) + 0\cdot 0 + 2\cdot 2 = 1.$

b) Second row of AB The second row of AB is the second row of A multiplied by B:

$$\begin{bmatrix}-1 & 0 & 2\end{bmatrix}\cdot B = \begin{bmatrix}5 & 16 & 2\end{bmatrix}$$

c) Third column of AB The third column of AB is A multiplied by the third column of B:

$$A\cdot\begin{bmatrix}0\\3\\1\end{bmatrix} = \begin{bmatrix}10\\2\\2\end{bmatrix}$$

6. Let the following matrices be defined:

$$A = \begin{bmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\\ d_1 & d_2 & d_3\end{bmatrix},\quad B = \begin{bmatrix}1 & 0 & 1 & 0\end{bmatrix},\quad C = \begin{bmatrix}1 & 0 & 1 & 0\\ 1 & -2 & 0 & 0\end{bmatrix},\quad D = \begin{bmatrix}1\\0\\1\end{bmatrix},\quad E = \begin{bmatrix}1 & 1 & 1\\ 0 & -2 & 0\\ 1 & 0 & 1\end{bmatrix}$$

Compute:

a) BA b) CA c) AD d) AE

Solution:

f) $B_{1\times 4}\cdot A_{4\times 3} \implies BA_{1\times 3} = \begin{bmatrix}a_1+c_1 & a_2+c_2 & a_3+c_3\end{bmatrix}$

f) $C_{2\times 4}\cdot A_{4\times 3} \implies CA_{2\times 3} = \begin{bmatrix}a_1+c_1 & a_2+c_2 & a_3+c_3\\ a_1-2b_1 & a_2-2b_2 & a_3-2b_3\end{bmatrix}$

f) $A_{4\times 3}\cdot D_{3\times 1} \implies AD_{4\times 1} = \begin{bmatrix}a_1+a_3\\ b_1+b_3\\ c_1+c_3\\ d_1+d_3\end{bmatrix}$

f) $A_{4\times 3}\cdot E_{3\times 3} \implies AE_{4\times 3} = \begin{bmatrix}a_1+a_3 & a_1-2a_2 & a_1+a_3\\ b_1+b_3 & b_1-2b_2 & b_1+b_3\\ c_1+c_3 & c_1-2c_2 & c_1+c_3\\ d_1+d_3 & d_1-2d_2 & d_1+d_3\end{bmatrix}$

7. Let the following matrices be defined:

$$A = \begin{bmatrix} 3 \\ 2 \\ -1 \\ 1 \end{bmatrix}, \qquad B = \begin{bmatrix} -1 & 0 & 1 & 2 \end{bmatrix}.$$

Compute AB and BA.

Solution:

Given $A \in \mathcal{M}_{4\times 1}(\mathbb{R})$ and $B \in \mathcal{M}_{1\times 4}(\mathbb{R})$, both products are defined.

$$AB = \begin{bmatrix} 3(-1) & 3(0) & 3(1) & 3(2) \\ 2(-1) & 2(0) & 2(1) & 2(2) \\ (-1)(-1) & (-1)(0) & (-1)(1) & (-1)(2) \\ 1(-1) & 1(0) & 1(1) & 1(2) \end{bmatrix} = \begin{bmatrix} -3 & 0 & 3 & 6 \\ -2 & 0 & 2 & 4 \\ 1 & 0 & -1 & -2 \\ -1 & 0 & 1 & 2 \end{bmatrix}.$$

$$\begin{aligned} BA &= \left[\sum_{k=1}^{4} b_{1k}a_{k1}\right] \\ &= \begin{bmatrix}(-1)(3) + 0(2) + 1(-1) + 2(1)\end{bmatrix} \\ &= \boxed{\begin{bmatrix}-2\end{bmatrix}}. \end{aligned}$$

8. Consider the matrices A and B from Exercise 5:

$$A = \begin{bmatrix} 0 & 3 & 1 \\ -1 & 0 & 2 \\ 3 & 0 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 3 \\ 2 & 8 & 1 \end{bmatrix}.$$

Determine all matrices X satisfying

$$2B^2 + \frac{1}{2}X - BA = (A - B)^2 + AB.$$

Solution:

Given $A, B \in \mathcal{M}_{3\times 3}(\mathbb{R})$, we seek $X \in \mathcal{M}_{3\times 3}(\mathbb{R})$.

Expanding $(A - B)^2$ without assuming $AB = BA$:

$$\begin{aligned} 2B^2 + \frac{1}{2}X - BA &= (A - B)^2 + AB \\ &= (A - B)(A - B) + AB \\ &= A^2 - AB - BA + B^2 + AB \\ &= A^2 - BA + B^2. \end{aligned}$$

Adding BA to both sides and isolating X:

$$
\begin{aligned}
2B^2 + \frac{1}{2}X &= A^2 + B^2 \\
\frac{1}{2}X &= A^2 + B^2 - 2B^2 \\
&= A^2 - B^2 \\
X &= 2(A^2 - B^2).
\end{aligned}
$$

Computing A^2 and B^2:

$$
\begin{aligned}
A^2 &= AA \\
&= \begin{bmatrix} 0 & 3 & 1 \\ -1 & 0 & 2 \\ 3 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 3 & 1 \\ -1 & 0 & 2 \\ 3 & 0 & 2 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & 8 \\ 6 & -3 & 3 \\ 6 & 9 & 7 \end{bmatrix}.
\end{aligned}
$$

$$
\begin{aligned}
B^2 &= BB \\
&= \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 3 \\ 2 & 8 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 3 \\ 2 & 8 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 6 & 25 & 0 \\ 0 & 0 & 25 \end{bmatrix}.
\end{aligned}
$$

Substituting:

$$
\begin{aligned}
X &= 2(A^2 - B^2) \\
&= 2\left(\begin{bmatrix} 0 & 0 & 8 \\ 6 & -3 & 3 \\ 6 & 9 & 7 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 6 & 25 & 0 \\ 0 & 0 & 25 \end{bmatrix} \right) \\
&= 2 \begin{bmatrix} -1 & 0 & 8 \\ 0 & -28 & 3 \\ 6 & 9 & -18 \end{bmatrix} \\
&= \boxed{\begin{bmatrix} -2 & 0 & 16 \\ 0 & -56 & 6 \\ 12 & 18 & -36 \end{bmatrix}}.
\end{aligned}
$$

Since the coefficient of X is nonzero, this solution is unique.

9. Let the following matrices be defined:

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & -1 \\ 2 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix}.$$

Compute AB, BA, and CA.
Solution:

Note: $A, B, C \in \mathcal{M}_{2\times 2}(\mathbb{R})$, so all three products are defined and belong to $\mathcal{M}_{2\times 2}(\mathbb{R})$.

Product AB:

$$AB = \begin{bmatrix} 4(-1)+2(2) & 4(-1)+2(2) \\ 2(-1)+1(2) & 2(-1)+1(2) \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}}.$$

Product BA:

$$BA = \begin{bmatrix} (-1)4+(-1)2 & (-1)2+(-1)1 \\ 2(4)+2(2) & 2(2)+2(1) \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} -6 & -3 \\ 12 & 6 \end{bmatrix}}.$$

Product CA:

$$CA = \begin{bmatrix} 0(4)+(-3)2 & 0(2)+(-3)1 \\ 3(4)+0(2) & 3(2)+0(1) \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} -6 & -3 \\ 12 & 6 \end{bmatrix}}.$$

10. Let

$$A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Compute A^2, A^3, A^4, and A^5.

Solution:

$$A^2 = A \cdot A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Since $A^2 = A$, it's clearer that, by the definition of the matrix power:

$$\forall p \in \mathbb{N}^+, \quad A^p = A^{p-1} \cdot A \implies$$

Therefore:

$$\implies A^3 = \underbrace{A^2}_{A} \cdot A = A \cdot A = A^2 = A$$
$$\implies A^4 = \underbrace{A^3}_{A} \cdot A = A \cdot A = A^2 = A$$
$$\implies A^5 = \underbrace{A^4}_{A} \cdot A = A \cdot A = A^2 = A$$

Alternative solution: (extra)

Since

$$\boxed{A^2 = A}$$

we may avoid the efford of calculte the other powers using induction.

This means:

$$\boxed{\forall n \in \mathbb{N}^+, \quad A^n = A}$$

Base case ($n = 1$):

$$A^1 = A.$$

Inductive hypothesis:

Let $n \in \mathbb{N}_{>0}$ and assume that $A^n = A$.

Inductive step:

Multiplying both sides of the inductive hypothesis by A on the right:

$$A^n = A \implies A^n A = AA.$$

Therefore,

$$\begin{aligned} A^{n+1} &= A^n A \\ &= AA \\ &= A^2 \\ &= A. \end{aligned}$$

Therefore, $P(n) \implies P(n+1) \implies \boxed{A^n = A}$

11. Consider the matrix

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}.$$

Verify that there exists a power of A which is the zero matrix.

Solution:

Given, $A \in \mathcal{M}_3(\mathbb{R})$ they are asking for:

$$\exists k \in \mathbb{N}+ : A^k = 0_3.$$

Computing A^2:

$$\begin{aligned} A^2 &= AA \\ &= \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \neq 0_3. \end{aligned}$$

Computing A^3:

$$\begin{aligned} A^3 &= A^2 A \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}. \end{aligned}$$

Therefore,

$$\boxed{\exists k = 3 \in \mathbb{N}^+ : A^k = 0_3}.$$

12. Consider the matrices

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Compute $(AB)^2$, A^2B^2, and B^2A^2.

Solution:

Given $A, B \in \mathcal{M}_{2\times 2}(\mathbb{R})$, all three products are defined.

Product $(AB)^2$:

First, compute AB:

$$AB = \begin{bmatrix} 1(1)+1(-1) & 1(1)+1(1) \\ 1(1)+0(-1) & 1(1)+0(1) \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix}$$

Therefore,

$$(AB)^2 = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} = \boxed{\begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}}$$

Product A^2B^2:

Computing both squares:

$$A^2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad B^2 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

Then,

$$A^2B^2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} = \boxed{\begin{bmatrix} -2 & 4 \\ -2 & 2 \end{bmatrix}}$$

Product B^2A^2:

$$B^2A^2 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \boxed{\begin{bmatrix} 2 & 2 \\ -4 & -2 \end{bmatrix}}$$

Note:

$$\boxed{(AB)^2 \neq A^2B^2 \neq B^2A^2}$$

In general, $(AB)^2 = ABAB$, but $ABAB \neq A^2B^2$ (without assuming $AB = BA$).

13. **State, with justification, whether the following statements are true or false.**

Let $n \in \mathbb{N}^+$ and $A, B, C \in \mathcal{M}_n(\mathbb{K})$.

a) **For any square matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ of the same order, $\boldsymbol{AB = BA}$.**
Universal statement:

$$\forall A, B \in \mathcal{M}_n(\mathbb{K}) : AB = BA.$$

False. Proof by counterexample.
Consider

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Then,

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = BA.$$

Therefore, the statement is false.

Sufficient condition:
If A and B are diagonal matrices, then, for all $i, j \in \{1, \ldots, n\}$,

$$(AB)_{ii} = a_{ii}b_{ii} = b_{ii}a_{ii} = (BA)_{ii},$$
$$(AB)_{ij} = 0 = (BA)_{ij}, \quad i \neq j.$$

Hence, all corresponding entries are equal, so

$$\boxed{AB = BA}.$$

Extra: $Otheroptionsare : A = 0, A = I_n, A = B, B = A^n$

b) **For any square matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ of the same order, if $\boldsymbol{AB = 0}$, then $\boldsymbol{A = 0}$ or $\boldsymbol{B = 0}$.**
Universal statement:

$$\forall A, B \in \mathcal{M}_n(\mathbb{K}) : AB = 0_n \implies (A = 0_n \vee B = 0_n).$$

False. Proof by counterexample.
Consider

$$A = B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq 0_2.$$

However,

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0_2$$

Thus,

$$AB = 0_2 \quad \wedge \quad A \neq 0_2 \quad \wedge \quad B \neq 0_2.$$

Therefore, the statement is false.

Sufficient condition:

If either A or B is invertible, the implication holds.

Case 1: A is invertible.

Multiplying by A^{-1} on the left:

$$\begin{aligned} AB = 0_n &\implies A^{-1}(AB) = A^{-1}0_n \\ &\implies (A^{-1}A)B = 0_n \\ &\implies I_nB = 0_n \\ &\implies \boxed{B = 0_n}. \end{aligned}$$

Case 2: B is invertible.

Multiplying by B^{-1} on the right:

$$\begin{aligned} AB = 0_n &\implies (AB)B^{-1} = 0_nB^{-1} \\ &\implies A(BB^{-1}) = 0_n \\ &\implies AI_n = 0_n \\ &\implies \boxed{A = 0_n}. \end{aligned}$$

Hence, if at least one factor is invertible, the other must be zero whenever $AB = 0_n$.

c) **For any square matrices A, B, and C of the same order, if $AB = AC$, then $B = C$.**

Universal statement:

$$\forall A, B, C \in \mathcal{M}_n(\mathbb{K}) : AB = AC \implies B = C.$$

False. Proof by counterexample.

Consider

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Then,

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = AC,$$

while $B \neq C$.

Therefore, the statement is false.

Sufficient condition:

If A is invertible, there exists $A^{-1} \in \mathcal{M}_n(\mathbb{K})$ such that

$$A^{-1}A = AA^{-1} = I_n.$$

Multiplying both sides of $AB = AC$ by A^{-1} on the left:

$$\begin{aligned} AB = AC &\implies A^{-1}(AB) = A^{-1}(AC) \\ &\implies (A^{-1}A)B = (A^{-1}A)C \\ &\implies I_n B = I_n C \\ &\implies \boxed{B = C}. \end{aligned}$$

Hence,

$$\boxed{A \text{ invertible} \wedge AB = AC \implies B = C.}$$

d) **If, for a square matrix A, there exists $k \in \mathbb{N}$ such that $A^k = 0$, then $A = 0$.**

Universal statement:

$$\forall A \in \mathcal{M}_n(\mathbb{K}) : \left(\exists k \in \mathbb{N}^+ : A^k = 0_n\right) \implies A = 0_n.$$

False. Proof by counterexample.

Consider

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq 0_2.$$

For $k = 2$,

$$A^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = 0_2$$

Thus,

$$\exists k = 2 \in \mathbb{N}^+ : A^k = 0_2, \quad A \neq 0_2.$$

Therefore, the statement is false.

Sufficient condition:

Suppose A is idempotent, i.e.,

$$A^2 = A.$$

By mathematical induction,

$$\forall p \in \mathbb{N}^+ : A^p = A.$$

Indeed, the base case $A^1 = A$ holds. Assuming $A^p = A$ for some $p \geq 1$, multiplying by A gives

$$\begin{aligned} A^{p+1} &= A^p A \\ &= AA \\ &= A^2 \\ &= A. \end{aligned}$$

Now, if

$$\exists k \in \mathbb{N}^+ : A^k = 0_n,$$

then, since $A^k = A$,

$$A = A^k = 0_n.$$

Therefore,

$$\boxed{A = 0_n}.$$

A matrix that is both idempotent and nilpotent must be zero.

e) **For any square matrices A and B of the same order, $(AB)^2 = A^2B^2$.**
Universal statement:

$$\forall A, B \in \mathcal{M}_n(\mathbb{K}) : (AB)^2 = A^2B^2.$$

False. Proof by counterexample.
Consider

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

Computing AB,

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Therefore,

$$(AB)^2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

However,

$$A^2 = 0_2, \qquad B^2 = 0_2,$$

so

$$A^2B^2 = 0_2.$$

Hence,

$$(AB)^2 \neq A^2B^2,$$

and the statement is false.

Sufficient condition:
Suppose $AB = BA$.
Then,

$$\begin{aligned}(AB)^2 &= (AB)(AB)\\ &= A(BA)B\\ &= A(AB)B\\ &= (AA)(BB)\\ &= \boxed{A^2B^2}.\end{aligned}$$

Therefore,

$$\boxed{AB = BA \implies (AB)^2 = A^2B^2.}$$

14. **Let $A_1, A_2, \ldots, A_n$ be matrices such that the product $A_1A_2\cdots A_n$ is defined.**

Express $(A_1A_2\cdots A_n)^T$ in terms of $A_1^T, A_2^T, \ldots, A_n^T$.

Using the transpose property

$$(AB)^T = B^TA^T,$$

and applying it successively, we obtain:

$$\begin{aligned}(A_1A_2\cdots A_n)^T &= ((A_1A_2\cdots A_{n-1})A_n)^T\\ &= A_n^T(A_1A_2\cdots A_{n-1})^T\\ &= A_n^TA_{n-1}^T(A_1A_2\cdots A_{n-2})^T\\ &\vdots\\ &= \boxed{A_n^TA_{n-1}^T\cdots A_2^TA_1^T}.\end{aligned}$$

So the transpose of a matrix product is the product of the transposed matrices in reverse order.

15. **Show that, for any square matrix A, the matrix $A + A^T$ is symmetric.**

By definition, a matrix A is symmetric if and only if $A = A^T$.

Let $A \in \mathcal{M}_n(\mathbb{K})$.
Note that $A^T \in \mathcal{M}_n(\mathbb{K})$, so $A + A^T$ is defined.

Using the properties of transposition:

$$\begin{aligned}(A + A^T)^T &= A^T + (A^T)^T\\ &= A^T + A\\ &= A + A^T \implies\\ \implies &\boxed{(A + A^T)^T = A + A^T}.\end{aligned}$$

16. **Indicate which of the following matrices are in echelon form and which are in reduced echelon form.**

a) $A = \begin{bmatrix} 1 & 1 & 5 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Echelon form, not reduced. The pivots move strictly to the right, but the second pivot is -1, not 1.

b) $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 5 & 1 & 1 \end{bmatrix}$

Neither. The first nonzero entries of the rows are in columns 1, 2 and 1, respectively.

c) $C = \begin{bmatrix} 1 & 5 & -4 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Neither. The second and third rows have their first nonzero entry in the same column.

d) $D = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

Echelon form, not reduced. The pivots are in positions $(1,1)$ and $(2,3)$, but $d_{13} = -4$ is nonzero above the second pivot.

e) $E = \begin{bmatrix} 1 & 5 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

Reduced echelon form. The pivots are 1, move strictly to the right and have only zeros above them. The zero row is at the bottom.

f) $F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

Reduced echelon form. The pivots are in positions $(1,1)$ and $(2,3)$, both equal 1, with zeros above them. The zero row is at the bottom.

g) $G = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Neither. The first row is zero, but there are nonzero rows below it.

h) $H = \begin{bmatrix} 0 & 2 & 4 & 0 \\ 0 & 0 & 8 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Echelon form, not reduced. The pivots are in positions $(1,2)$ and $(2,3)$, and the zero rows are at the bottom. However, the pivots are 2 and 8, not 1.

i) $I = \begin{bmatrix} 1 & 5 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Neither. The second row is zero, but the third is not.

j) $J = \begin{bmatrix} 1 & 5 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Reduced echelon form. There is only one pivot, equal to 1, and all remaining rows are zero.

k) $K = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Reduced echelon form. The pivots are in positions $(1,1)$ and $(2,4)$, both equal 1, with zeros above them. The zero rows are at the bottom.

l) $L = \begin{bmatrix} 0 & 1 & -7 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Reduced echelon form. The only pivot is $l_{12} = 1$, and all remaining rows are zero.

m) $M = \begin{bmatrix} 0 & 1 & -7 \end{bmatrix}$

Reduced echelon form. The only pivot is $m_{12} = 1$. The remaining conditions hold trivially.

n) $N = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

Neither. The first two rows have their first nonzero entry in the same column.

o) $O = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Reduced echelon form. All rows are zero. Since there are no pivots, the conditions concerning them hold trivially.

17. **For each of the following matrices:**

- **Obtain a row-equivalent matrix in echelon form using elementary row operations.**
- **Determine its rank.**
- **Obtain a row-equivalent matrix in reduced echelon form using elementary row operations.**

a)

$$A = \begin{bmatrix} 2 & 4 & 0 & 3 \\ 0 & 0 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ -1 & 4 & 0 & 4 \end{bmatrix}$$

b)

$$B = \begin{bmatrix} 0 & 0 & 1 \\ 4 & 1 & 0 \\ 1 & 2 & -1 \end{bmatrix}$$

c)

$$C = \begin{bmatrix} 0 & 0 & 3 & 3i & -15 \\ i & 3 & 2 & 0 & 3 \\ 2 & -6i & -5i & 1 & 2 \end{bmatrix}$$

d)

$$D = \begin{bmatrix} 2 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 \\ 4 & -4 & 4 & 4 \\ 0 & 2 & 0 & 0 \end{bmatrix}$$

e)

$$E = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 2 & 1 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

f)

$$F = \begin{bmatrix} 3 & 0 & 0 & -21 & 24 \\ 4 & 1 & 0 & -25 & 34 \\ -2 & 3 & 0 & 22 & -2 \end{bmatrix}$$

18. **Consider the following matrices.**

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 1 & 1 \\ -1 & 1 & 2 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 & 1 & 1 \\ -2 & -4 & -6 & -2 \\ -2 & -1 & -1 & 0 \end{bmatrix}$$

a) Determine elementary matrices H, J, K such that $HA = B$, $JA = C$ and $KA = D$.

Let $T_{jk}(c)$ be I_3 with entry (j,k) replaced by c, for $j \neq k$. Left multiplication by $T_{jk}(c)$ performs $F_j \leftarrow F_j + cF_k$. Let S_{23} interchange rows $2, 3$ and let $D_j(c)$ multiply row j by c. These are elementary matrices when $c \neq 0$ in $D_j(c)$. Here

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix}.$$

B is obtained by interchanging rows $1, 2$; C by $F_2 \leftarrow F_2 + F_3$; D by $F_2 \leftarrow -2F_2$. Therefore

$$H = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad J = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Direct multiplication gives $HA = B$, $JA = C$, $KA = D$.

b) Determine, if they exist, elementary matrices $E_p, \ldots, E_1$ such that

i)
$$E_p \cdots E_1 A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 2 & 2 & 0 \end{bmatrix}.$$

Apply $F_2 \leftarrow F_2 - F_1$, $F_3 \leftarrow F_3 + 2F_1$ and $F_2 \leftrightarrow F_3$. Thus

$$S_{23}T_{31}(2)T_{21}(-1)A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 2 & 2 & 0 \end{bmatrix}.$$

Take $E_1 = T_{21}(-1)$, $E_2 = T_{31}(2)$, $E_3 = S_{23}$.

ii)
$$E_p \cdots E_1 A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 3 & -3 & -6 \\ 0 & 0 & 7 & 7 \end{bmatrix}.$$

Continue from (i) with $F_3 \leftarrow F_3 + 2F_2$, $F_2 \leftarrow -3F_2$, $F_3 \leftarrow \frac{7}{4}F_3$. Hence take

$$E_1 = T_{21}(-1), \quad E_2 = T_{31}(2), \quad E_3 = S_{23},$$
$$E_4 = T_{32}(2), \quad E_5 = D_2(-3), \quad E_6 = D_3(7/4).$$

Indeed,
$$E_6 \cdots E_1 A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 3 & -3 & -6 \\ 0 & 0 & 7 & 7 \end{bmatrix}.$$

The handwritten resolution prints 0 in position $(1,4)$ of its final matrix. The target in the exercise sheet and the stated operations give 1 there.

c) Determine a matrix Q such that $QA = F$, where

$$F = \begin{bmatrix} -2 & 4 & 2 & -2 \\ 0 & 2 & 2 & 0 \\ 0 & 3 & 9 & 6 \end{bmatrix}.$$

Apply to $[A \mid I_3]$ the operations

$$F_2 \leftarrow F_2 - F_1, \quad F_3 \leftarrow F_3 + 2F_1, \quad F_1 \leftarrow F_1 - F_2,$$
$$F_3 \leftarrow F_3 + F_2, \quad F_1 \leftarrow -2F_1, \quad F_3 \leftarrow 3F_3.$$

The result is
$$\left[\begin{array}{cccc|ccc} -2 & 4 & 2 & -2 & -4 & 2 & 0 \\ 0 & 2 & 2 & 0 & -1 & 1 & 0 \\ 0 & 3 & 9 & 6 & 3 & 3 & 3 \end{array}\right].$$

Consequently $Q = \begin{bmatrix} -4 & 2 & 0 \\ -1 & 1 & 0 \\ 3 & 3 & 3 \end{bmatrix}$ satisfies $QA = F$.

19. **Discuss the rank of each of the following matrices in terms of the real parameters α and β occurring in it.**

a) $A = \begin{bmatrix} -5 & 1 & 8 \\ 0 & \alpha & 0 \\ 0 & 0 & \alpha - 1 \end{bmatrix}$

$A = \begin{bmatrix} -5 & 1 & 8 \\ 0 & \alpha & 0 \\ 0 & 0 & \alpha - 1 \end{bmatrix}$. If $\alpha \notin \{0, 1\}$, there are three pivots. For $\alpha = 0$ or 1, exactly two rows remain independent. Thus

$$\mathrm{r}(A) = \begin{cases} 2, & \alpha \in \{0, 1\}, \\ 3, & \alpha \notin \{0, 1\}. \end{cases}$$

b) $B = \begin{bmatrix} 4 & \alpha - 1 & 3 \\ 0 & \alpha^2 + 1 & \alpha - 1 \\ 0 & 0 & \alpha + 5 \end{bmatrix}$

$B = \begin{bmatrix} 4 & \alpha - 1 & 3 \\ 0 & \alpha^2 + 1 & \alpha - 1 \\ 0 & 0 & \alpha + 5 \end{bmatrix}$. Since $4 \neq 0$ and $\alpha^2 + 1 > 0$, the first two pivots always exist. Therefore $\mathrm{r}(B) = 2$ for $\alpha = -5$, and $\mathrm{r}(B) = 3$ otherwise.

c) $C = \begin{bmatrix} -1 & 1 & 3 \\ 1 & \alpha - 1 & -\alpha - 3 \\ -1 - \alpha & 2 - \alpha & 3\alpha + 5 \end{bmatrix}$

For $C = \begin{bmatrix} -1 & 1 & 3 \\ 1 & \alpha - 1 & -\alpha - 3 \\ -1 - \alpha & 2 - \alpha & 3\alpha + 5 \end{bmatrix}$, apply $F_2 \leftarrow F_2 + F_1$, $F_3 \leftarrow F_3 - (1+\alpha)F_1$, $F_3 \leftarrow F_3 + 2F_2$, $F_2 \leftrightarrow F_3$, $F_3 \leftarrow F_3 - \alpha F_2$. This gives

$$C \sim \begin{bmatrix} -1 & 1 & 3 \\ 0 & 1 & 2 - 2\alpha \\ 0 & 0 & \alpha(2\alpha - 3) \end{bmatrix}.$$

No division by a parameter was used. Hence $\mathrm{r}(C) = 2$ for $\alpha \in \{0, \frac{3}{2}\}$, and $\mathrm{r}(C) = 3$ otherwise.

d) $D = \begin{bmatrix} 1 & 2 & -1 & 2 \\ 3 & 2\alpha + 6 & \beta^2 - 4 & \beta^2 + 9 \\ -1 & -2 & \beta^2 & \beta - 1 \end{bmatrix}$

For

$$D = \begin{bmatrix} 1 & 2 & -1 & 2 \\ 3 & 2\alpha + 6 & \beta^2 - 4 & \beta^2 + 9 \\ -1 & -2 & \beta^2 & \beta - 1 \end{bmatrix},$$

$F_2 \leftarrow F_2 - 3F_1$, $F_3 \leftarrow F_3 + F_1$ give

$$D \sim \begin{bmatrix} 1 & 2 & -1 & 2 \\ 0 & 2\alpha & \beta^2 - 1 & \beta^2 + 3 \\ 0 & 0 & \beta^2 - 1 & \beta + 1 \end{bmatrix}.$$

If $\beta = -1$, the third row is zero and the second is nonzero. If $\beta = 1$, the last two rows are independent exactly when $\alpha \neq 0$. If $\beta \notin \{-1, 1\}$ and $\alpha \neq 0$, there are three pivots. If $\beta \notin \{-1, 1\}$ and $\alpha = 0$, subtracting row 3 from row

2 produces the nonzero last entry $\beta^2-\beta+2=(\beta-\frac{1}{2})^2+\frac{7}{4}$; the last two rows are again independent. Therefore

$$\mathrm{r}(D)=\begin{cases}2, & \beta=-1,\text{ or }(\beta=1\text{ and }\alpha=0),\\ 3, & \text{otherwise.}\end{cases}$$

20. **Consider the matrices**

$$A=\begin{bmatrix}3&5\\1&2\end{bmatrix},\quad B=\begin{bmatrix}2&-5\\-1&3\end{bmatrix},\quad C=\begin{bmatrix}0&-1\\1&-1\end{bmatrix}.$$

a) Verify that B is the inverse of A.

Here $A=\begin{bmatrix}3&5\\1&2\end{bmatrix}$, $B=\begin{bmatrix}2&-5\\-1&3\end{bmatrix}$, $C=\begin{bmatrix}0&-1\\1&-1\end{bmatrix}$.

$$AB=\begin{bmatrix}6-5&-15+15\\2-2&-5+6\end{bmatrix}=I_2,\quad BA=\begin{bmatrix}6-5&10-10\\-3+3&-5+6\end{bmatrix}=I_2.$$

Thus $B=A^{-1}$.

b) Is the matrix $\begin{bmatrix}2&-5\\-1&2\end{bmatrix}$ the inverse of A? Justify your answer.

For $D=\begin{bmatrix}2&-5\\-1&2\end{bmatrix}$, $AD=\begin{bmatrix}1&-5\\0&-1\end{bmatrix}\neq I_2$, so $D\neq A^{-1}$.

c) Determine $(5A)^{-1}$ and $(A^T)^{-1}$.

$$(5A)^{-1}=\tfrac{1}{5}B=\begin{bmatrix}2/5&-1\\-1/5&3/5\end{bmatrix},\qquad (A^T)^{-1}=B^T=\begin{bmatrix}2&-1\\-5&3\end{bmatrix}.$$

d) Compute C^3. Is C invertible? What is its inverse?

$$C^2=\begin{bmatrix}-1&1\\-1&0\end{bmatrix},\quad C^3=I_2.$$

Therefore $C^{-1}=C^2$, since $CC^2=C^2C=I_2$.

e) Is AC invertible? Justify your answer and, if it is, determine $(AC)^{-1}$.

The product is invertible, with

$$(AC)^{-1}=C^{-1}A^{-1}=C^2B=\begin{bmatrix}-3&8\\-2&5\end{bmatrix}.$$

Indeed $(AC)(C^{-1}A^{-1})=I_2=(C^{-1}A^{-1})(AC)$.

21. **Consider two diagonal matrices of order n:**

$$A=\begin{bmatrix}a_1&0&\cdots&0\\0&a_2&\cdots&0\\\vdots&\vdots&\ddots&\vdots\\0&0&\cdots&a_n\end{bmatrix},\quad B=\begin{bmatrix}b_1&0&\cdots&0\\0&b_2&\cdots&0\\\vdots&\vdots&\ddots&\vdots\\0&0&\cdots&b_n\end{bmatrix}.$$

a) Compute AB.

If $A = \operatorname{diag}(a_1, \ldots, a_n)$ and $B = \operatorname{diag}(b_1, \ldots, b_n)$, then $(AB)_{ij} = \sum_k a_{ik} b_{kj}$ vanishes when $i \neq j$ and equals $a_i b_i$ when $i = j$. Thus $AB = \operatorname{diag}(a_1 b_1, \ldots, a_n b_n)$.

b) What conditions must $a_1, a_2, \ldots, a_n$ satisfy for A to be invertible? When A is invertible, what is A^{-1}?

A is invertible exactly when every $a_j \neq 0$, and then

$$A^{-1} = \operatorname{diag}(a_1^{-1}, \ldots, a_n^{-1}).$$

Both products with A equal I_n. If $a_j = 0$, row j of AX is zero for every X, so AX cannot equal I_n.

22. **Let $A_1, A_2, \ldots, A_n$ be invertible square matrices of the same order. Express $(A_1 A_2 \cdots A_n)^{-1}$ in terms of $A_1^{-1}, A_2^{-1}, \ldots, A_n^{-1}$.**

$$(A_1 A_2 \cdots A_n)^{-1} = A_n^{-1} A_{n-1}^{-1} \cdots A_1^{-1}.$$

In either order of multiplication, associativity allows adjacent factors $A_j A_j^{-1}$ or $A_j^{-1} A_j$ to cancel successively, leaving the identity.

23. **Show that if $A \in M_n(\mathbb{K})$ is invertible, then $\overline{A}$ is also invertible and $(\overline{A})^{-1} = \overline{A^{-1}}$.**

Conjugating $AA^{-1} = A^{-1}A = I_n$ gives

$$\overline{A}\,\overline{A^{-1}} = I_n = \overline{A^{-1}}\,\overline{A}.$$

Hence $\overline{A}$ is invertible and $(\overline{A})^{-1} = \overline{A^{-1}}$.

24. **Let $A \in M_n(\mathbb{K})$ satisfy $A^3 + A = 5I_n$.**

a) Justify that A is invertible and determine A^{-1} in terms of A.

Assume $A^3 + A = 5I_n$.

Since

$$A\frac{A^2 + I_n}{5} = \frac{A^3 + A}{5} = I_n = \frac{A^2 + I_n}{5}A,$$

A is invertible and $A^{-1} = \frac{1}{5}(A^2 + I_n)$.

b) Justify that $7A$ is invertible and determine $(7A)^{-1}$ in terms of A.

As $7 \neq 0$, $7A$ is invertible and $(7A)^{-1} = \frac{1}{7}A^{-1} = \frac{1}{35}(A^2 + I_n)$.

25. **Let $A \in M_n(\mathbb{K})$ be invertible. Show that**

a) if $B \in M_{n\times m}(\mathbb{K})$ satisfies $AB = 0_{n\times m}$, then $B = 0_{n\times m}$.

$AB = 0_{n\times m}$ implies $B = A^{-1}(AB) = A^{-1}0_{n\times m} = 0_{n\times m}$.

b) if $B \in M_{m\times n}(\mathbb{K})$ satisfies $BA = 0_{m\times n}$, then $B = 0_{m\times n}$.

$BA = 0_{m\times n}$ implies $B = (BA)A^{-1} = 0_{m\times n}A^{-1} = 0_{m\times n}$.

c) if $B, C \in M_{n\times m}(\mathbb{K})$ satisfy $AB = AC$, then $B = C$.

$AB = AC$ implies $A^{-1}AB = A^{-1}AC$, hence $B = C$.

d) if $B, C \in M_{m\times n}(\mathbb{K})$ satisfy $BA = CA$, then $B = C$.
$BA = CA$ implies $BAA^{-1} = CAA^{-1}$, hence $B = C$.
All products have the dimensions specified in the question.

26. **For each of the following matrices, determine whether it is invertible and, if it is, compute its inverse.**

a) $A = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}$
For the row reductions below, apply the stated operations to $[M \mid I]$. When the left block becomes I, the right block is M^{-1}.
$A = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}$ is not invertible: $F_2 \leftarrow F_2 - 4F_1$ produces a zero row, so $\mathrm{r}(A) = 1 < 2$.

b) $B = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$
For $B = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$, use $F_2 \leftarrow F_2 - F_1$, $F_2 \leftarrow -\frac{1}{3}F_2$, $F_1 \leftarrow F_1 - F_2$. Then

$$B^{-1} = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix}.$$

c) $C = \begin{bmatrix} 5 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 7 \end{bmatrix}$
$C = \mathrm{diag}(5, -4, 7)$ is invertible. Dividing its three rows by $5, -4, 7$ gives $C^{-1} = \mathrm{diag}(\frac{1}{5}, -\frac{1}{4}, \frac{1}{7})$.

d) $D = \begin{bmatrix} 0 & 0 & 5 \\ 0 & -4 & 0 \\ 7 & 0 & 0 \end{bmatrix}$
For $D = \begin{bmatrix} 0 & 0 & 5 \\ 0 & -4 & 0 \\ 7 & 0 & 0 \end{bmatrix}$, interchange rows $1, 3$ and divide the resulting rows by $7, -4, 5$. Thus

$$D^{-1} = \begin{bmatrix} 0 & 0 & 1/7 \\ 0 & -1/4 & 0 \\ 1/5 & 0 & 0 \end{bmatrix}.$$

e) $E = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
$E = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is invertible. The operation $F_1 \leftarrow F_1 - 4F_3$ gives $E^{-1} = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

f) $F = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$F = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ satisfies $F^2 = I_3$, so $F^{-1} = F$.

g) $G = \begin{bmatrix} 0 & 3 & 1 \\ -1 & 3 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

For $G = \begin{bmatrix} 0 & 3 & 1 \\ -1 & 3 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, use

$$F_1 \leftrightarrow F_2, \quad F_1 \leftarrow -F_1, \quad F_2 \leftrightarrow F_3, \quad F_3 \leftarrow F_3 - 3F_2, \quad F_3 \leftarrow -\tfrac{1}{2}F_3,$$
$$F_2 \leftarrow F_2 - F_3, \quad F_1 \leftarrow F_1 + F_3, \quad F_1 \leftarrow F_1 + 3F_2.$$

The left block becomes I_3, and

$$G^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 1/2 & 0 & -1/2 \\ -1/2 & 0 & 3/2 \end{bmatrix}.$$

h) $H = \begin{bmatrix} 2 & -1 & 4 \\ 1 & -1 & 2 \\ -1 & 0 & -2 \end{bmatrix}$

$H = \begin{bmatrix} 2 & -1 & 4 \\ 1 & -1 & 2 \\ -1 & 0 & -2 \end{bmatrix}$ is not invertible. Its third column is twice its first column, so $H \begin{bmatrix} -2 & 0 & 1 \end{bmatrix}^T = 0$ with a nonzero column. If H^{-1} existed, multiplication by it would give a contradiction.

i) $I = \begin{bmatrix} 2 & -1 & 4 \\ 1 & -1 & 2 \\ -1 & 0 & -3 \end{bmatrix}$

For $I = \begin{bmatrix} 2 & -1 & 4 \\ 1 & -1 & 2 \\ -1 & 0 & -3 \end{bmatrix}$, use

$$F_1 \leftarrow \tfrac{1}{2}F_1, \quad F_2 \leftarrow F_2 - F_1, \quad F_3 \leftarrow F_3 + F_1, \quad F_2 \leftarrow -2F_2,$$
$$F_3 \leftarrow F_3 + \tfrac{1}{2}F_2, \quad F_3 \leftarrow -F_3, \quad F_1 \leftarrow F_1 - 2F_3, \quad F_1 \leftarrow F_1 + \tfrac{1}{2}F_2.$$

Hence I is invertible and $I^{-1} = \begin{bmatrix} 3 & -3 & 2 \\ 1 & -2 & 0 \\ -1 & 1 & -1 \end{bmatrix}$.

j) $J = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & -1 & -1 \end{bmatrix}$

$J = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & -1 & -1 \end{bmatrix}$ is not invertible: its first row is the sum of rows $2, 3$.

Equivalently, $J \begin{bmatrix} 1 & -3 & 5 \end{bmatrix}^T = 0$ with a nonzero column.

k) $K = \begin{bmatrix} 1 & -1 & 1 & 2 \\ 2 & -2 & 1 & 1 \\ 1 & -1 & 0 & 1 \\ -2 & 0 & 2 & -2 \end{bmatrix}$

For $K = \begin{bmatrix} 1 & -1 & 1 & 2 \\ 2 & -2 & 1 & 1 \\ 1 & -1 & 0 & 1 \\ -2 & 0 & 2 & -2 \end{bmatrix}$, apply

$$F_2 \leftarrow F_2 - 2F_1, \quad F_3 \leftarrow F_3 - F_1, \quad F_4 \leftarrow F_4 + 2F_1, \quad F_2 \leftrightarrow F_4,$$
$$F_2 \leftarrow -\tfrac{1}{2}F_2, \quad F_3 \leftarrow -F_3, \quad F_4 \leftarrow F_4 + F_3, \quad F_4 \leftarrow -\tfrac{1}{2}F_4.$$

The augmented matrix is now

$$\left[\begin{array}{cccc|cccc} 1 & -1 & 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & -1 & -1 & 0 & 0 & -1/2 \\ 0 & 0 & 1 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1/2 & -1/2 & 1/2 & 0 \end{array}\right].$$

Complete the reduction with

$$F_1 \leftarrow F_1 - 2F_4, \quad F_2 \leftarrow F_2 + F_4, \quad F_3 \leftarrow F_3 - F_4,$$
$$F_1 \leftarrow F_1 - F_3, \quad F_2 \leftarrow F_2 + 2F_3, \quad F_1 \leftarrow F_1 + F_2.$$

Thus

$$K^{-1} = \begin{bmatrix} 0 & 1 & -2 & -1/2 \\ 1/2 & 1/2 & -5/2 & -1/2 \\ 1/2 & 1/2 & -3/2 & 0 \\ 1/2 & -1/2 & 1/2 & 0 \end{bmatrix}.$$

27. **For each $\alpha \in \mathbb{R}$, consider the matrix**

$$A_\alpha = \begin{bmatrix} 1 & \alpha & \alpha & \alpha \\ 0 & 0 & \alpha+3 & \alpha+2 \\ 1 & \alpha & \alpha & -1 \\ -1 & 1-\alpha & 1 & 7 \end{bmatrix}.$$

a) Determine the values of α for which A_α is invertible.

$$A_\alpha = \begin{bmatrix} 1 & \alpha & \alpha & \alpha \\ 0 & 0 & \alpha+3 & \alpha+2 \\ 1 & \alpha & \alpha & -1 \\ -1 & 1-\alpha & 1 & 7 \end{bmatrix}.$$

Apply $F_3 \leftarrow F_3 - F_1$, $F_4 \leftarrow F_4 + F_1$, $F_2 \leftrightarrow F_4$, $F_3 \leftrightarrow F_4$. Then

$$A_\alpha \sim \begin{bmatrix} 1 & \alpha & \alpha & \alpha \\ 0 & 1 & \alpha+1 & \alpha+7 \\ 0 & 0 & \alpha+3 & \alpha+2 \\ 0 & 0 & 0 & -\alpha-1 \end{bmatrix}.$$

For $\alpha \notin \{-3,-1\}$, there are four pivots. For $\alpha = -1$ the last row is zero. For $\alpha = -3$, the last two rows have nonzero entries only in column 4 and are dependent. Therefore

$$A_\alpha \text{ is invertible} \iff \alpha \in \mathbb{R} \setminus \{-3,-1\}.$$

b) Compute the inverse of A_0.

For $\alpha = 0$, apply the same operations to $[A_0 \mid I_4]$:

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 7 & 1 & 0 & 0 & 1 \\ 0 & 0 & 3 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 1 & 0 \end{array}\right].$$

Now use $F_4 \leftarrow -F_4$, $F_3 \leftarrow F_3 - 2F_4$, $F_3 \leftarrow \frac{1}{3}F_3$, $F_2 \leftarrow F_2 - 7F_4$, $F_2 \leftarrow F_2 - F_3$. The left block is I_4, giving

$$A_0^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -16/3 & -1/3 & 19/3 & 1 \\ -2/3 & 1/3 & 2/3 & 0 \\ 1 & 0 & -1 & 0 \end{bmatrix}.$$

28. **Compute the inverse of the matrix**

$$A = \begin{bmatrix} 1 & 0 & i \\ 0 & 2i & -1 \\ 2 & -4 & i \end{bmatrix}.$$

For $A = \begin{bmatrix} 1 & 0 & i \\ 0 & 2i & -1 \\ 2 & -4 & i \end{bmatrix}$, apply to $[A \mid I_3]$ the operations $F_3 \leftarrow F_3 - 2F_1$, $F_2 \leftarrow -\frac{i}{2}F_2$, $F_3 \leftarrow F_3 + 4F_2$, $F_3 \leftarrow -iF_3$. This gives

$$\left[\begin{array}{ccc|ccc} 1 & 0 & i & 1 & 0 & 0 \\ 0 & 1 & i/2 & 0 & -i/2 & 0 \\ 0 & 0 & 1 & 2i & -2 & -i \end{array}\right].$$

Finally, $F_1 \leftarrow F_1 - iF_3$ and $F_2 \leftarrow F_2 - \frac{i}{2}F_3$ give

$$A^{-1} = \begin{bmatrix} 3 & 2i & -1 \\ 1 & i/2 & -1/2 \\ 2i & -2 & -i \end{bmatrix}.$$

29. **Let**

$$A = \begin{bmatrix} 3 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ -1 & 0 \end{bmatrix}.$$

a) Compute the inverse of B.

Here

$$A = \begin{bmatrix} 3 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ -1 & 0 \end{bmatrix}.$$

On $[B \mid I_3]$, use

$$F_2 \leftarrow -\tfrac{1}{2}F_2, \quad F_3 \leftarrow F_3 - F_2, \quad F_3 \leftarrow \tfrac{2}{3}F_3, \quad F_2 \leftarrow F_2 + \tfrac{1}{2}F_3, \quad F_1 \leftarrow F_1 - 2F_2.$$

Then B is invertible and

$$B^{-1} = \begin{bmatrix} 1 & 2/3 & -2/3 \\ 0 & -1/3 & 1/3 \\ 0 & 1/3 & 2/3 \end{bmatrix}.$$

b) Solve $AX = 2X + BC + C$ for $X \in M_{3\times 2}(\mathbb{R})$.

Since $A - 2I_3 = B$, the equation is equivalent to $BX = BC + C$. Multiplying on the left by B^{-1} gives the unique solution

$$X = C + B^{-1}C = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ -1 & 0 \end{bmatrix} + \begin{bmatrix} 5/3 & 4/3 \\ -1/3 & -2/3 \\ -2/3 & 2/3 \end{bmatrix} = \begin{bmatrix} 8/3 & 4/3 \\ -1/3 & 4/3 \\ -5/3 & 2/3 \end{bmatrix}.$$

Substitution gives $BX = BC + C$, hence the original equation.

30. **Let $\boldsymbol{A} \in \boldsymbol{M_{m\times n}}(\mathbb{K})$ and let $\boldsymbol{P} \in \boldsymbol{M_m}(\mathbb{K})$ be invertible. Show that $\mathbf{r}(\boldsymbol{PA}) = \mathbf{r}(\boldsymbol{A})$.**

Since P is invertible, row reduction takes it to I_m: $E_s \cdots E_1 P = I_m$, where the E_j are elementary matrices. Thus $P = E_1^{-1} \cdots E_s^{-1}$. Each inverse is elementary, so PA is obtained from A by elementary row operations. Such operations preserve rank; therefore $\mathrm{r}(PA) = \mathrm{r}(A)$.

31. **Determine whether each of the following equations is linear in the unknowns $\boldsymbol{x, y, z}$.**

a) $x + \pi y + \sqrt[3]{2}\, z = e$.

An equation is linear in x, y, z if it has the form $ax + by + cz = d$ with fixed coefficients and no products or nonlinear functions of the unknowns.

$x + \pi y + \sqrt[3]{2}\, z = e$: linear; all coefficients are constants.

b) $x^2 + xy + z = 2$.

$x^2 + xy + z = 2$: not linear, because of x^2 and xy.

c) $x + \sin y + 2z = 0$.
$x + \sin y + 2z = 0$: not linear, because of $\sin y$.

d) $4x + (\sin \frac{\pi}{10})^2 y + 23z = \cos \frac{\pi}{5}$.
$4x + (\sin \frac{\pi}{10})^2 y + 23z = \cos \frac{\pi}{5}$: linear; the trigonometric expressions and 23 are constants.

e) $(-x + 8y - 2z)^2 = 1$.
$(-x + 8y - 2z)^2 = 1$: not linear. It is equivalent to the union of the two equations $-x + 8y - 2z = 1$ and $-x + 8y - 2z = -1$, not to a single linear equation.

f) $ax + 6y - 3z = 9$, with $a \in \mathbb{R}$ fixed.
$ax + 6y - 3z = 9$, with fixed $a \in \mathbb{R}$: linear, including when $a = 0$.

32. **Solve the following systems by Gauss–Jordan elimination. Then give three particular solutions and the general solution of the associated homogeneous system.**

a)
$$\begin{cases} x_1 - 3x_2 + 2x_3 + 2x_4 + 7x_5 = 8, \\ x_1 - 3x_2 + 2x_3 + 3x_4 + 11x_5 = 11, \\ 3x_1 - 9x_2 + 6x_3 - 3x_4 - 15x_5 = -3, \\ -2x_1 + 6x_2 - 4x_3 + 2x_4 + 10x_5 = 2. \end{cases}$$

The unknowns are x_1, x_2, x_3, x_4, x_5.
The augmented matrix is

$$\left[\begin{array}{rrrrr|r} 1 & -3 & 2 & 2 & 7 & 8 \\ 1 & -3 & 2 & 3 & 11 & 11 \\ 3 & -9 & 6 & -3 & -15 & -3 \\ -2 & 6 & -4 & 2 & 10 & 2 \end{array}\right].$$

Apply
$$F_2 \leftarrow F_2 - F_1, \quad F_3 \leftarrow F_3 - 3F_1, \quad F_4 \leftarrow F_4 + 2F_1,$$
$$F_3 \leftarrow F_3 + 9F_2, \quad F_4 \leftarrow F_4 - 6F_2, \quad F_1 \leftarrow F_1 - 2F_2.$$
The reduced augmented matrix is

$$\left[\begin{array}{rrrrr|r} 1 & -3 & 2 & 0 & -1 & 2 \\ 0 & 0 & 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Put $x_2 = s$, $x_3 = t$, $x_5 = u$. The general solution is
$$(x_1, x_2, x_3, x_4, x_5) = (2, 0, 0, 3, 0) + s(3, 1, 0, 0, 0) + t(-2, 0, 1, 0, 0) + u(1, 0, 0, -4, 1).$$
There are three free unknowns. Three particular solutions are $(2, 0, 0, 3, 0)$, $(5, 1, 0, 3, 0)$ and $(0, 0, 1, 3, 0)$. The general solution of the associated homogeneous system is
$$s(3, 1, 0, 0, 0) + t(-2, 0, 1, 0, 0) + u(1, 0, 0, -4, 1), \qquad s, t, u \in \mathbb{R}.$$

b)

$$\begin{cases} 3x_1 + 7x_3 + 5x_4 = 21, \\ x_1 + 3x_3 + 3x_4 = 9, \\ 2x_1 + 5x_3 + 4x_4 = 15, \\ x_1 + 2x_3 + x_4 = 6. \end{cases}$$

The unknowns are x_1, x_2, x_3, x_4.

The augmented matrix is

$$\left[\begin{array}{cccc|c} 3 & 0 & 7 & 5 & 21 \\ 1 & 0 & 3 & 3 & 9 \\ 2 & 0 & 5 & 4 & 15 \\ 1 & 0 & 2 & 1 & 6 \end{array}\right].$$

Apply

$$F_1 \leftrightarrow F_2, \quad F_2 \leftarrow F_2 - 3F_1, \quad F_3 \leftarrow F_3 - 2F_1, \quad F_4 \leftarrow F_4 - F_1,$$
$$F_2 \leftarrow -\tfrac{1}{2}F_2, \quad F_3 \leftarrow F_3 + F_2, \quad F_4 \leftarrow F_4 + F_2, \quad F_1 \leftarrow F_1 - 3F_2.$$

This gives

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -3 & 0 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Putting $x_4 = s$, $x_2 = t$, the general solution is

$$(x_1, x_2, x_3, x_4) = (0, 0, 3, 0) + s(3, 0, -2, 1) + t(0, 1, 0, 0).$$

Three particular solutions are $(0, 0, 3, 0)$, $(0, 1, 3, 0)$ and $(3, 0, 1, 1)$. The associated homogeneous system has general solution $s(3, 0, -2, 1) + t(0, 1, 0, 0)$, with $s, t \in \mathbb{R}$.

33. **State, with justification, whether each of the following assertions is true or false for any linear system (S) with m equations and n unknowns.**

a) If $m > n$, then (S) is inconsistent.
False. The system $x = 1$, $2x = 2$ has $m = 2 > 1 = n$ and a unique solution.

b) If $m < n$, then (S) is inconsistent.
False. The equation $x + y = 0$ has $m = 1 < 2 = n$ and has solutions.

c) If $m < n$, then (S) is either inconsistent or has infinitely many solutions.
True. If the system is consistent, its number of pivots is at most $m < n$, so at least one unknown is free. Thus it has infinitely many solutions. Otherwise it is inconsistent.

d) If (S) has a unique solution, then $m = n$.
False. The counterexample in (a) has a unique solution with $m \neq n$.

e) If $m = n$, then (S) has a unique solution.
False. The system $x + y = 0$, $2x + 2y = 0$ has $m = n = 2$ and infinitely many solutions. Also, $x + y = 0$, $x + y = 1$ is inconsistent with $m = n = 2$.

34. **Discuss the following systems in the unknowns x, y, z in terms of the parameter a. When there are infinitely many solutions, state the degree of indeterminacy.**

a)

$$\begin{cases} x + y - az = 0, \\ ax + y - z = 1 - a, \\ x + ay - z = 1 - a. \end{cases}$$

The system is

$$x + y - az = 0, \qquad ax + y - z = 1 - a, \qquad x + ay - z = 1 - a.$$

Subtract the first equation from the second and third:

$$(a-1)(x+z) = 1 - a, \qquad (a-1)(y+z) = 1 - a.$$

For $a = 1$, all three original equations reduce to $x + y - z = 0$. The general solution is $(x, y, z) = (s, t, s + t)$; the degree of indeterminacy is 2.
For $a \neq 1$, $x + z = y + z = -1$. Substitution in the first equation gives $(a + 2)z = -2$. If $a = -2$, this is impossible. If $a \notin \{-2, 1\}$, the unique solution is

$$(x, y, z) = \left(-\frac{a}{a+2}, -\frac{a}{a+2}, -\frac{2}{a+2} \right).$$

b)

$$\begin{cases} x + y - z = 0, \\ ay + (2a - 1)z = a, \\ ax + 2ay = 0, \\ ax + ay + (1 - 2a)z = a - 1. \end{cases}$$

The four equations are

$$x + y - z = 0, \qquad ay + (2a - 1)z = a,$$
$$ax + 2ay = 0, \quad ax + ay + (1 - 2a)z = a - 1.$$

The left side of equation 2 plus equation 4 minus equation 3 is zero; the same combination on the right is $2a - 1$. Consistency therefore requires $a = \frac{1}{2}$. At this value, equation 2 gives $y = 1$, equation 3 gives $x = -2$, and equation 1 gives $z = -1$. These values satisfy equation 4. Thus the system is uniquely solvable for $a = \frac{1}{2}$, with solution $(-2, 1, -1)$, and is inconsistent for every other a. There is no indeterminate case.

35. **For each $a, b \in \mathbb{R}$, consider the following linear system in the unknowns x, y, z:**

$$(S_{a,b}): \quad \begin{cases} -x + ay + bz = b, \\ x - y + z = 2, \\ x - ay + z = 1. \end{cases}$$

a) Discuss $(S_{a,b})$ in terms of a and b.

$$(S_{a,b}): \quad -x+ay+bz=b, \qquad x-y+z=2, \qquad x-ay+z=1.$$

Add equations $1, 3$, and subtract equation 3 from equation 2. Together with equation 2, this gives the equivalent system

$$(b+1)z=b+1, \qquad (a-1)y=1, \qquad x-y+z=2.$$

If $a=1$, the system is inconsistent. If $a\neq 1$ and $b\neq -1$, it has the unique solution

$$(x,y,z)=\left(\frac{a}{a-1},\frac{1}{a-1},1\right).$$

If $a\neq 1$ and $b=-1$, its general solution is

$$(x,y,z)=\left(2+\frac{1}{a-1}-t,\frac{1}{a-1},t\right), \qquad t\in\mathbb{R}.$$

The degree of indeterminacy is 1.

b) Give the values of a, b for which the associated homogeneous system has a nonzero solution.

The associated homogeneous system is equivalent to

$$(b+1)z=0, \qquad (a-1)y=0, \qquad x-y+z=0.$$

It has a nonzero solution exactly when $a=1$ or $b=-1$. Indeed, when $a=1$ take $(1,1,0)$, and when $b=-1$ take $(-1,0,1)$. If neither equality holds, $z=y=x=0$.

c) Give the values of a, b for which the coefficient matrix is invertible.

The coefficient matrix is invertible exactly when its homogeneous system has only the zero solution, namely $a\neq 1$ and $b\neq -1$.

d) Solve $(S_{-3,-1})$ by Gauss–Jordan elimination.

For $(a,b)=(-3,-1)$, start with

$$\left[\begin{array}{ccc|c} -1 & -3 & -1 & -1 \\ 1 & -1 & 1 & 2 \\ 1 & 3 & 1 & 1 \end{array}\right].$$

Apply $F_1\leftrightarrow F_2$, $F_2\leftarrow F_2+F_1$, $F_3\leftarrow F_3-F_1$, $F_2\leftarrow -\frac{1}{4}F_2$, $F_3\leftarrow F_3-4F_2$, $F_1\leftarrow F_1+F_2$. This gives

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 7/4 \\ 0 & 1 & 0 & -1/4 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Therefore $(x,y,z)=(\frac{7}{4}-t,-\frac{1}{4},t)$, $t\in\mathbb{R}$.

36. **Let**

$$A=\begin{bmatrix} 1 & 2 & 2 & -1 \\ 2 & 4 & -2 & 2 \\ 0 & 0 & 6 & -4 \end{bmatrix}, \qquad B=\begin{bmatrix} -1 \\ 4 \\ -6 \end{bmatrix}.$$

Let (S) be the system in the unknowns x_1, x_2, x_3, x_4 with matrix form $AX=B$.

a) Without solving (S), show that $(2,0,-3,-3)$ is a solution and that $(2,0,0,-3)$ is not.

$$A=\begin{bmatrix}1&2&2&-1\\2&4&-2&2\\0&0&6&-4\end{bmatrix},\qquad B=\begin{bmatrix}-1\\4\\-6\end{bmatrix}.$$

Direct multiplication gives

$$A\begin{bmatrix}2\\0\\-3\\-3\end{bmatrix}=\begin{bmatrix}2-6+3\\4+6-6\\-18+12\end{bmatrix}=B,\qquad A\begin{bmatrix}2\\0\\0\\-3\end{bmatrix}=\begin{bmatrix}5\\-2\\12\end{bmatrix}\neq B.$$

b) Solve the associated homogeneous system by Gauss–Jordan elimination.
For $AX=0$, apply $F_2\leftarrow F_2-2F_1$, $F_3\leftarrow F_3+F_2$, $F_2\leftarrow -\frac{1}{6}F_2$, $F_1\leftarrow F_1-2F_2$:

$$A\sim\begin{bmatrix}1&2&0&1/3\\0&0&1&-2/3\\0&0&0&0\end{bmatrix}.$$

Put $x_2=s$, $x_4=t$. The homogeneous solutions are

$$(x_1,x_2,x_3,x_4)=s(-2,1,0,0)+t\left(-\tfrac{1}{3},0,\tfrac{2}{3},1\right).$$

c) Give the general solution of (S).
Adding the particular solution from (a) gives all solutions:

$$(x_1,x_2,x_3,x_4)=(2,0,-3,-3)+s(-2,1,0,0)+t\left(-\tfrac{1}{3},0,\tfrac{2}{3},1\right),\qquad s,t\in\mathbb{R}.$$

This is exhaustive because the difference of any two solutions satisfies $AX=0$.

d) Give two solutions of (S) distinct from the solution considered in (a).
Taking $(s,t)=(1,0)$ and $(0,3)$ gives $(0,1,-3,-3)$ and $(1,0,-1,0)$, both distinct from $(2,0,-3,-3)$.

37. **Consider the following linear system in the unknowns x,y,z,w:**

$$(S):\quad\begin{cases}x+2y-2z-w=0,\\2x+y-z+4w=0,\\-x-y+z-w=0.\end{cases}$$

Give a linear system whose associated homogeneous system is (S) and which has $(2,-1,0,-1)$ as a solution.

Keep the given left sides and evaluate them at $(2,-1,0,-1)$:

$$2-2+1=1,\qquad 4-1-4=-1,\qquad -2+1+1=0.$$

A suitable system is therefore

$$\begin{cases} x + 2y - 2z - w = 1, \\ 2x + y - z + 4w = -1, \\ -x - y + z - w = 0. \end{cases}$$

Its associated homogeneous system is exactly the given system.

38. **Let (S) be a linear system over $\mathbb{R}$ in the unknowns x, y, z, w, with general solution**

$$(x, y, z, w) = (2, 1, -1, 0) + \alpha(1, 3, 0, 1) + \beta(1, 0, 0, 1), \qquad \alpha, \beta \in \mathbb{R}.$$

a) Justify that (S) is not homogeneous.

Write $p = (2, 1, -1, 0)$, $u = (1, 3, 0, 1)$, $v = (1, 0, 0, 1)$. The given general solution is $p + \alpha u + \beta v$. Every such point satisfies $z = -1$ and $x - w = 2$. Conversely, any point satisfying these two equations has the given form: choose $\alpha = (y - 1)/3$ and $\beta = w - \alpha$.

The zero vector is not a solution, since its third coordinate is not -1. Every homogeneous system has the zero solution, so this system is not homogeneous.

b) State, with justification, which of the following families in $\mathbb{R}^4$ are general solutions of (S):

i)

$$(x, y, z, w) = (3, 1, -1, 1) + \alpha(1, 3, 0, 1) + \beta(1, 0, 0, 1), \qquad \alpha, \beta \in \mathbb{R}.$$

Yes. $(3, 1, -1, 1) = p + v$. Replacing the particular solution by this point amounts to replacing β by $\beta + 1$, which still runs through all real values.

ii)

$$(x, y, z, w) = (1, 4, -1, -1) + \alpha(1, 3, 0, 1) + \beta(1, 0, 0, 1), \qquad \alpha, \beta \in \mathbb{R}.$$

Yes. $(1, 4, -1, -1) = p + u - 2v$. Translating the two parameters by $(1, -2)$ does not change the set of solutions.

iii)

$$(x, y, z, w) = (3, 1, -1, 0) + \alpha(1, 3, 0, 1) + \beta(1, 0, 0, 1), \qquad \alpha, \beta \in \mathbb{R}.$$

No. $(3, 1, -1, 0)$ has $x - w = 3$, so the proposed family already contains a nonsolution at $\alpha = \beta = 0$.

c) Justify that the family

$$(x, y, z, w) = (2, 1, -1, 0) + \alpha(3, 4, -1, 1) + \beta(1, 0, 0, 1), \qquad \alpha, \beta \in \mathbb{R}.$$

is not a general solution of (S).

In the proposed family the third coordinate is $-1 - \alpha$. For example, $\alpha = 1$, $\beta = 0$ gives $(5, 5, -2, 1)$, which is not a solution. Hence that family cannot be a general solution of the given system.

39. **For $a, b \in \mathbb{R}$, consider the matrices**

$$A = \begin{bmatrix} 8 & -2 & 1 & 3 \\ 0 & a^2-4 & a-2 & 6 \\ 0 & 0 & a^2-4 & b \\ 0 & 0 & 0 & a \end{bmatrix}, \qquad B = \begin{bmatrix} b \\ 1 \\ 2 \\ 2b \end{bmatrix}.$$

a) Let (S) be the linear system with augmented matrix $[A \mid B]$. Discuss (S) in terms of a and b.

The coefficient and right-hand-side matrices are

$$A = \begin{bmatrix} 8 & -2 & 1 & 3 \\ 0 & a^2-4 & a-2 & 6 \\ 0 & 0 & a^2-4 & b \\ 0 & 0 & 0 & a \end{bmatrix}, \qquad B = \begin{bmatrix} b \\ 1 \\ 2 \\ 2b \end{bmatrix}.$$

Use the equations from bottom to top.

- If $a \notin \{-2, 0, 2\}$, all four diagonal entries are nonzero, so the system has a unique solution. Explicitly, writing the unknowns as x, y, z, w,

$$w = \frac{2b}{a}, \quad z = \frac{2-bw}{a^2-4}, \quad y = \frac{1-(a-2)z-6w}{a^2-4}, \quad x = \frac{b+2y-z-3w}{8}.$$

- If $a = 0$, the last equation requires $b = 0$. For $b \neq 0$ the system is inconsistent. For $b = 0$, the remaining equations give

$$(x, y, z, w) = \left(\tfrac{1}{16}, \tfrac{3}{2}t, -\tfrac{1}{2}, t\right), \qquad t \in \mathbb{R}.$$

Its degree of indeterminacy is 1.

- If $a = 2$, row 2 gives $w = 1/6$. Row 4 gives $w = b$, so $b = 1/6$; row 3 gives $bw = 2$, which would instead require $b = 12$. The system is inconsistent.
- If $a = -2$, row 4 gives $w = -b$ and row 3 gives $bw = 2$. Thus $-b^2 = 2$, impossible for real b. The system is inconsistent.

b) Determine the values of a, b for which $A - 5I_4$ is invertible.

The upper triangular matrix $A - 5I_4$ has diagonal entries $3, a^2-9, a^2-9, a-5$. It is invertible exactly when all are nonzero:

$$a \notin \{-3, 3, 5\}, \qquad b \in \mathbb{R} \text{ arbitrary}.$$

c) For the values found in (b), state, with justification, whether there exists a nonzero column matrix $C \in M_{4\times 1}(\mathbb{R})$ such that $AC = 5C$.

For those values, $AC = 5C$ is equivalent to $(A - 5I_4)C = 0$. Multiplication by $(A - 5I_4)^{-1}$ gives $C = 0$. There is no nonzero such column.

40. **Let $A \in M_n(\mathbb{C})$ satisfy $AA^* = 3I_n$.**

a) Show that A is invertible and determine its inverse in terms of A.

Assume $A \in M_n(\mathbb{C})$ and $AA^* = 3I_n$.

$A(\frac{1}{3}A^*) = I_n$. A square matrix with a right inverse is invertible; hence $A^{-1} = \frac{1}{3}A^*$. For completeness, the right inverse gives a solution of $AX = Y$ for every column Y, so the echelon form of A has a pivot in every row. Since A is square it reduces to I_n, proving invertibility.

b) Justify that $\overline{A}$ is invertible and determine its inverse in terms of A.

By Exercise 23, $\overline{A}$ is invertible and

$$(\overline{A})^{-1} = \overline{A^{-1}} = \tfrac{1}{3}\overline{A^*} = \tfrac{1}{3}A^T.$$